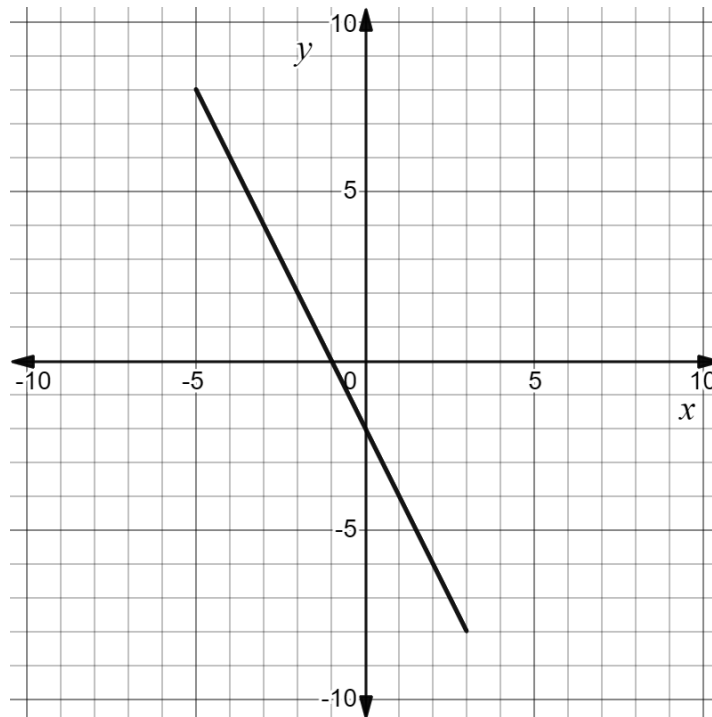


Algebra 1
Unit 4
Lesson 1 Practice

Name _____
Date _____
Period _____

The graph of $y = -\frac{7}{3}x - 2$ is shown. The graph has a restricted domain.



1. Give a written description for the range.
2. Write the domain using inequalities.
3. Write the domain using set-builder notation.
4. Write the range using set-builder notation.

For each written description complete the table.

Written Description	Inequalities	Set-builder notation
5. All values less than or equal to $5\frac{3}{4}$		
6. All values greater than -2.6 but less than or equal to 8.9		
7. All values greater than $-\frac{22}{6}$		

8. Write a written description of the set of values represented by $\{x: 30.2 > x \geq -3.8\}$.

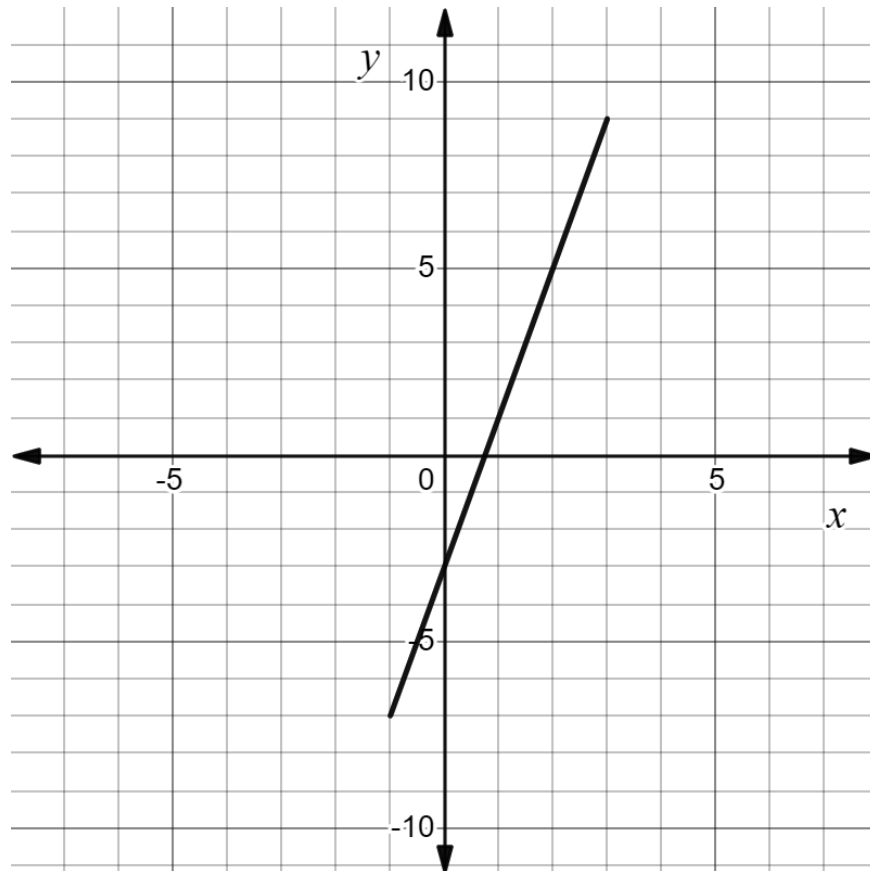
9. Write the set of values represented by $\{h| h < -0.25\}$ using inequalities.

10. Write the set of values represented by $c \geq \frac{5}{6}$ using set-builder notation.

Algebra 1
Unit 4
Lesson 2 Practice

Name _____
Date _____
Period _____

The graph of linear function is shown. The graph has a restricted domain.



1. The slope of the line segment is _____.
2. As the x -values are increasing, the y -values are _____.
3. The coordinates of the point where the line crosses the y -axis are _____.
The y -value of the point is _____.
4. The coordinates of the point where the line crosses the x -axis are _____.
The x -value of the point is _____.

Identify the key features of the horizontal line $y = -3$ in questions 5-6.

5.

Domain	Range	As x -values increase, y -values ...

6.

Slope	x -intercept	y -intercept

Identify the key features of the line $y = \frac{2}{3}x$ in questions 7-8.

7.

Domain	Range	As x -values increase, y -values ...

8.

Slope	x -intercept	y -intercept

Identify the key features of the line $y = -2x + 13$ in questions 9-10.

9.

Domain	Range	As x -values increase, y -values ...

10.

Slope	x -intercept	y -intercept

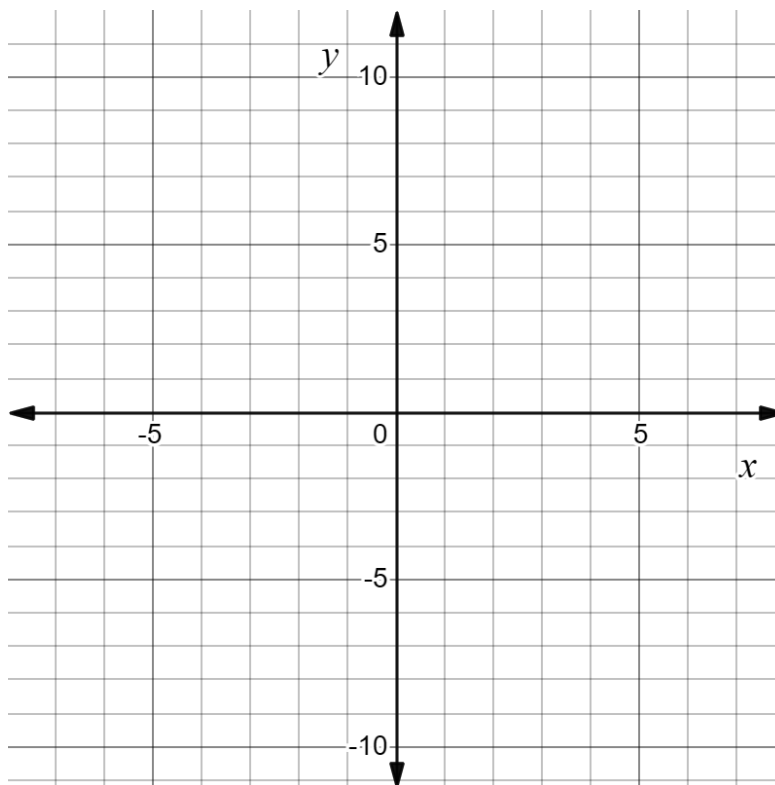
Algebra 1
Unit 4
Lesson 3 Practice

Name _____
Date _____
Period _____

1. Complete the table of values for $y = -\frac{3}{4}x - 3$.

x	-8	-4	0	4	8
y					

2. Graph the equation $y = -\frac{3}{4}x - 3$.



3. Write the domain of the line $y = -\frac{3}{4}x - 3$ using set-builder notation.

4. What is the x -intercept of the line $y = -\frac{3}{4}x - 3$?

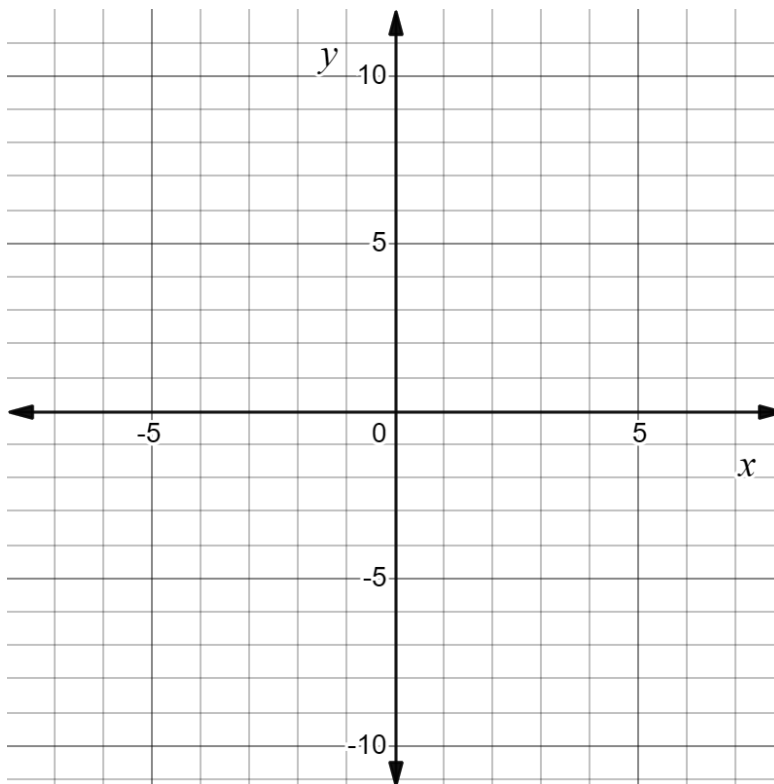
5. Complete the statement about the line $y = -\frac{3}{4}x - 3$.

As x -values increase, y -values _____

6. Complete the table of values for $y + 5 = -4(x - 2)$.

x	-2	-1	0	1	2
y					

7. Graph the equation $y + 5 = -4(x - 2)$.



8. Use inequalities to complete the table for the line $y + 5 = -4(x - 2)$.

Domain	Range

9. What is the y -intercept of the line $y + 5 = -4(x - 2)$?

10. Complete the statement about the line $y - 3 = 3(x + 1)$.

As x -values increase, y -values _____

Algebra 1
Unit 4
Lesson 4 Practice

Name _____
Date _____
Period _____

A study of the average emissions of carbon dioxide (CO_2) from a vehicle in the Netherlands was conducted from 2000 to 2019. The function $C(x) = -5x + 190$ measures the carbon intensity of newly registered passenger vehicles in the Netherlands where x is the number of years since 2000 and C is the average emissions of CO_2 (in grams) per kilometer.

1. Two students were discussing how to find the domain of the function $C(x) = -5x + 190$. Their explanations are shown.

Michelle	Jaheim
The domain is all real numbers because the function is linear.	The domain should start at 0 because x represents the number of years since 2000. It would make sense for the domain to end with 19 because that is the last year of the study.

Determine which student is correct and write the domain using inequalities.

2. Write the range of the function $C(x) = -5x + 190$ using inequalities.
3. Determine the x -intercept.
4. Interpret the x -intercept.
5. Determine the y -intercept.
6. Interpret the y -intercept.

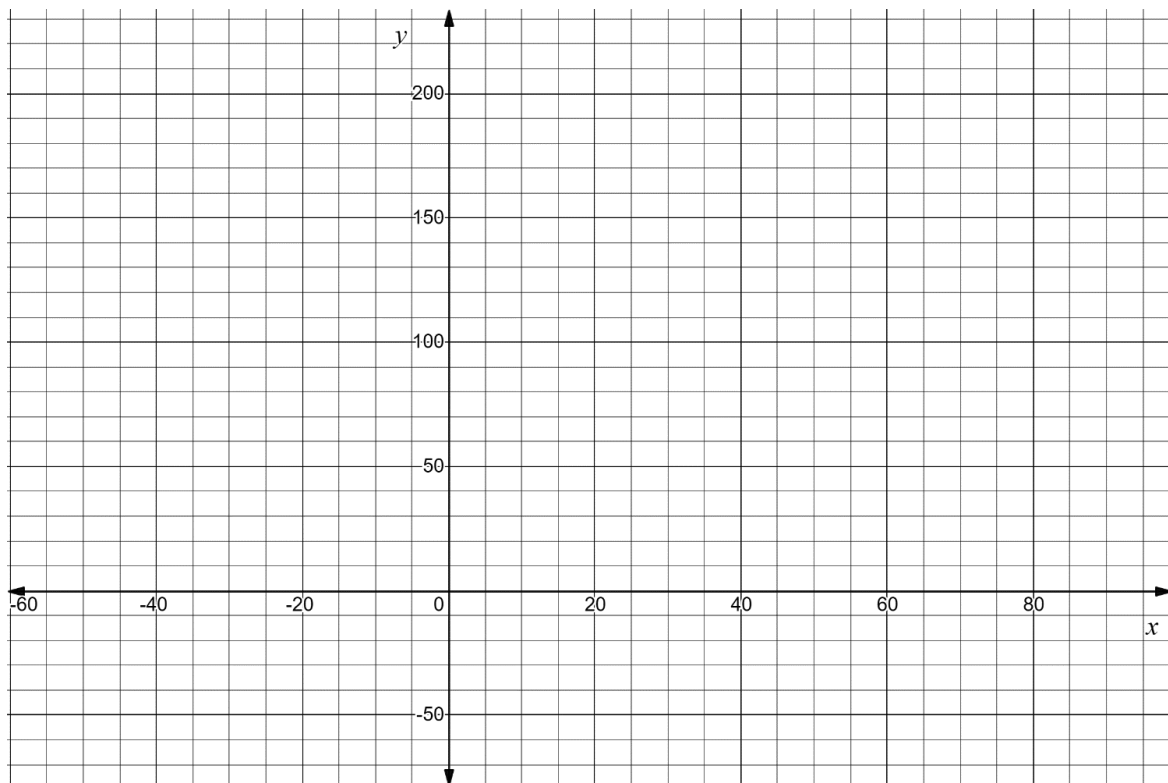
7. Complete the statement about the line $C(x) = -5x + 190$.

As x -values increase, y -values _____

8. Determine the slope.

9. Interpret the slope.

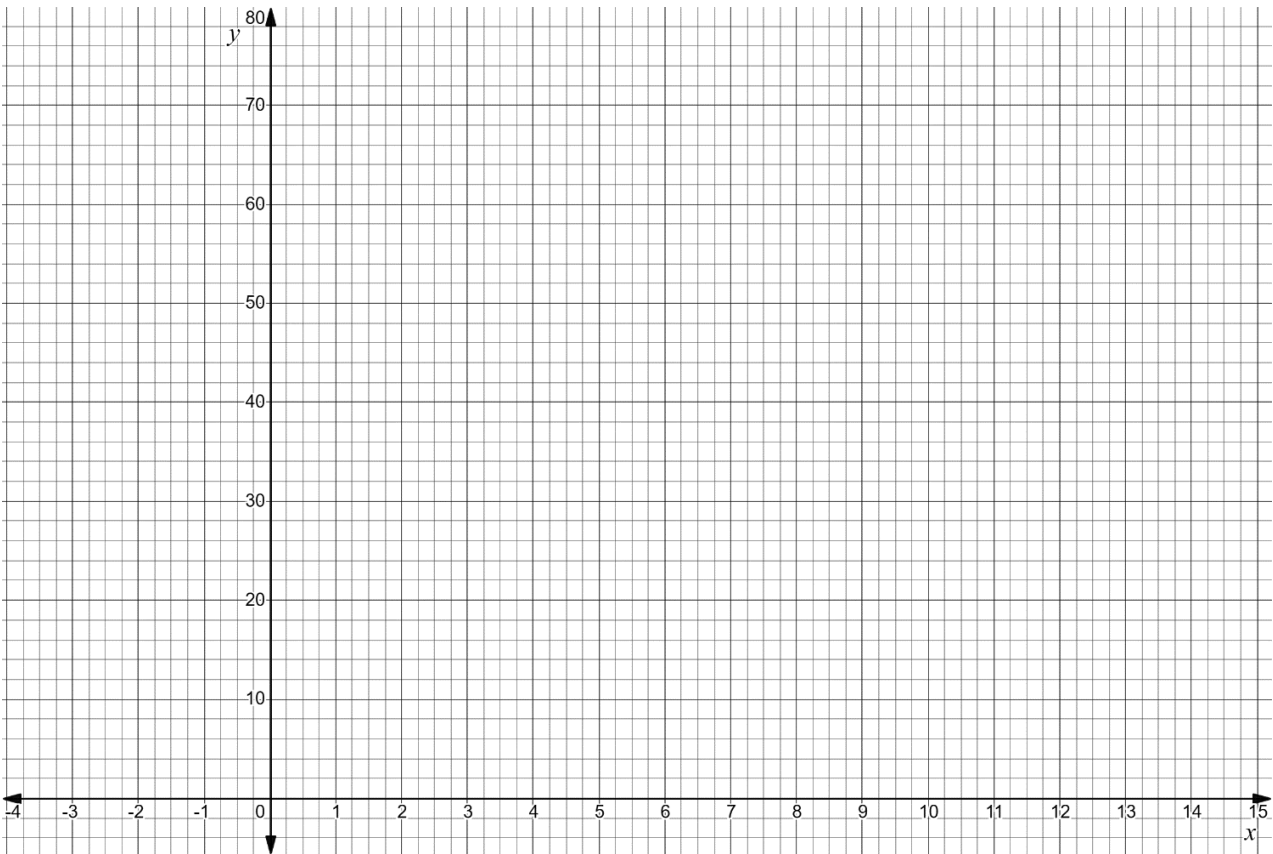
10. Graph the function.



The table of values shows the linear relationship between x , the height of water in a container, and y , the weight of the container.

Height of Water (inches)	2	4.5	5	7.25	8
Weight of Container (ounces)	27	57	63	90	99

1. Graph the linear relationship.



2. Place a check in the box of a key feature that can be determined from the table of values.

Domain	Range	x -intercept	y -intercept	Slope	Increasing/ Decreasing

3. The equation $12x - y = -3$ models the linear relationship. Place a check in the box of a key feature that can be determined using the equation.

Domain	Range	x -intercept	y -intercept	Slope	Increasing/ Decreasing

4. Write the equation $12x - y = -3$ in slope-intercept form.

5. What domain makes sense for the context?

6. What range makes sense for the context?

7. Determine the x -intercept.

8. Interpret the x -intercept.

9. Determine the slope.

10. Interpret the slope.

For each real-world representation of a context write an inequality that represents the constraint.

1. A person has to be at least 18 years old to vote.
2. A job offers a salary between \$42,000 and \$50,000.
3. The highest altitude for skydiving without oxygen is 14,000 feet.

An automatic soap dispenser holds 18 ounces (oz) of soap. Each time someone places their hand under the soap dispenser releases 0.15 oz of soap. The function $f(x) = 18 - 0.15x$ models the amount of soap, in ounces, in the dispenser for x dispenses.

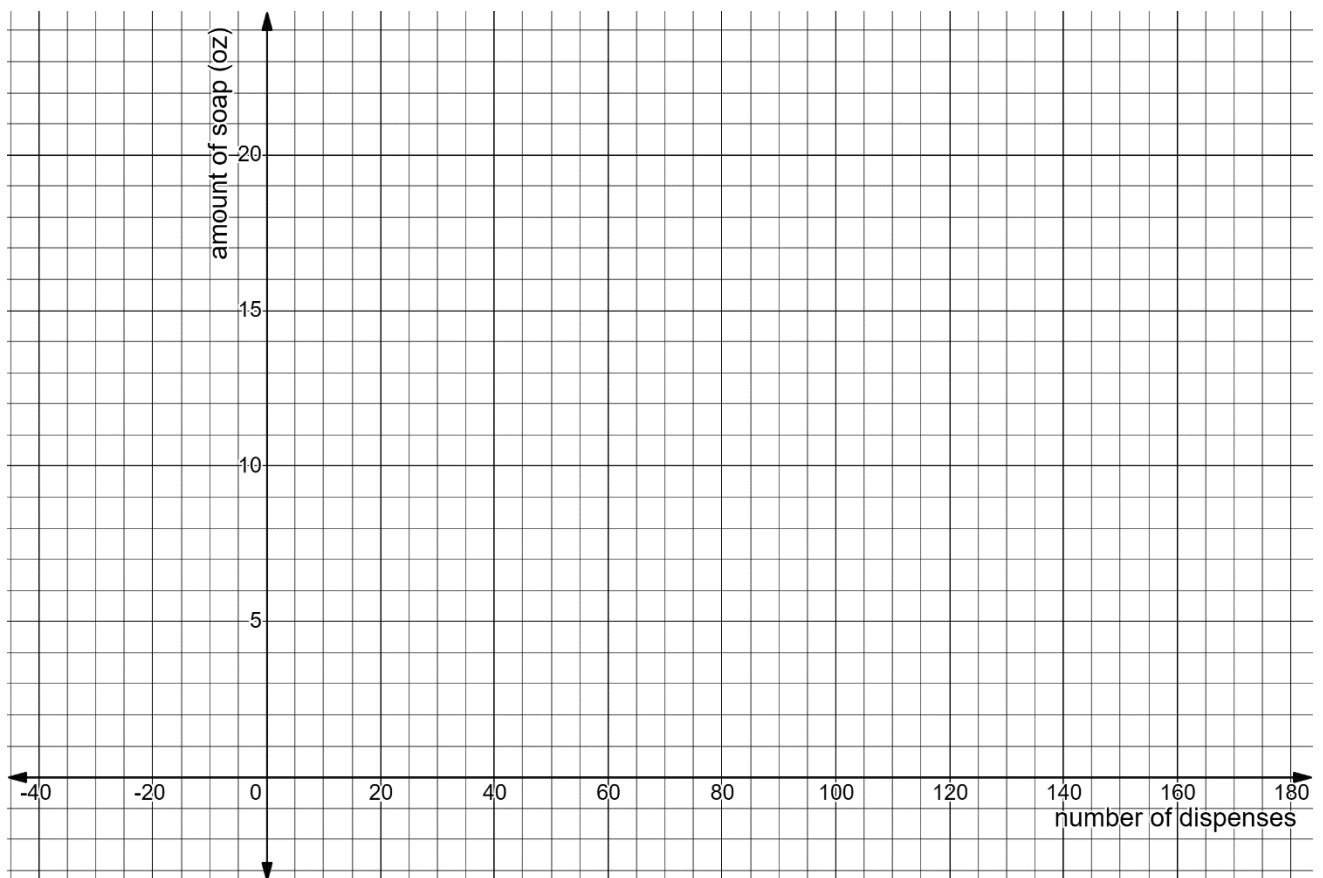
4. Find the value of x for which $f(x) = 13.2$.
5. In terms of the context, what does the value of x that is the solution to $f(x) = 13.2$ mean?
6. Find the x -intercept.
7. What does the x -intercept mean within the context?

8. What key feature is $f(0)$?

9. Select all of the constraints for the context.

- ☐ $x \geq 0$
- ☐ $x \leq 18$
- ☐ $x \leq 120$
- ☐ $f \geq 0$
- ☐ $f \leq 18$
- ☐ $f \leq 120$

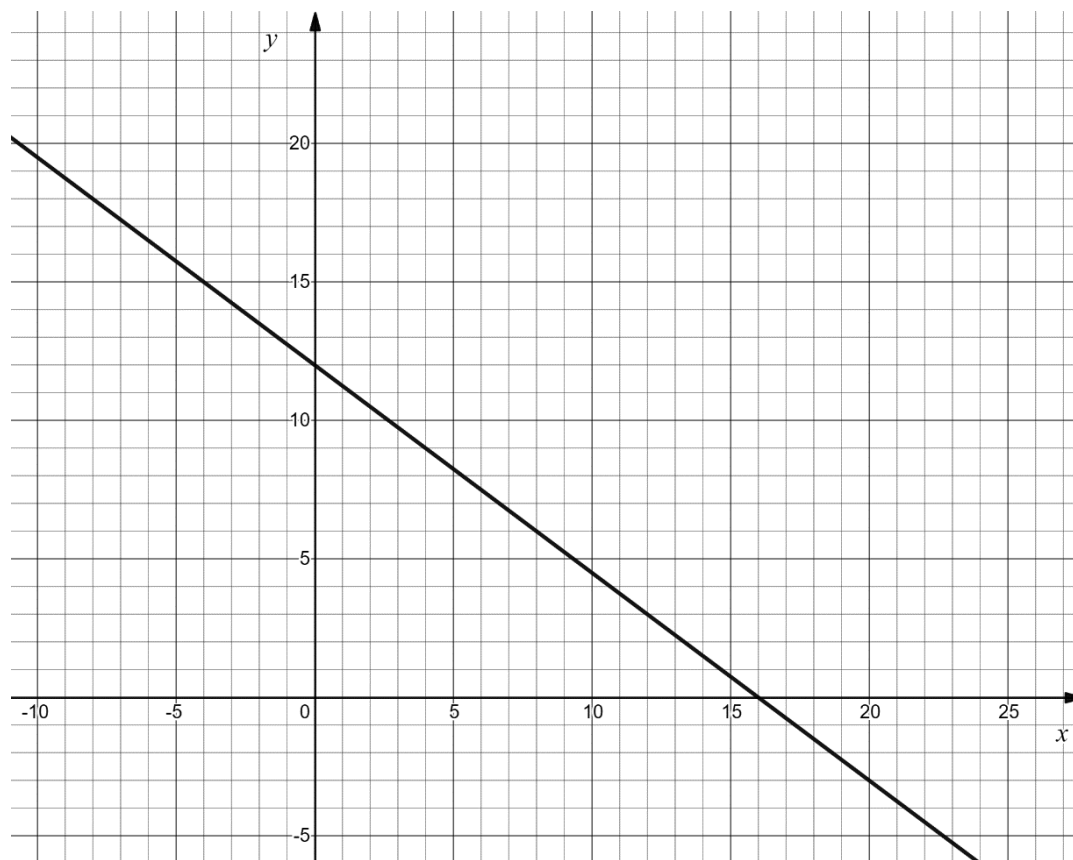
10. Graph the function in terms of the context.



Algebra 1
Unit 4
Lesson 7 Practice

Name _____
Date _____
Period _____

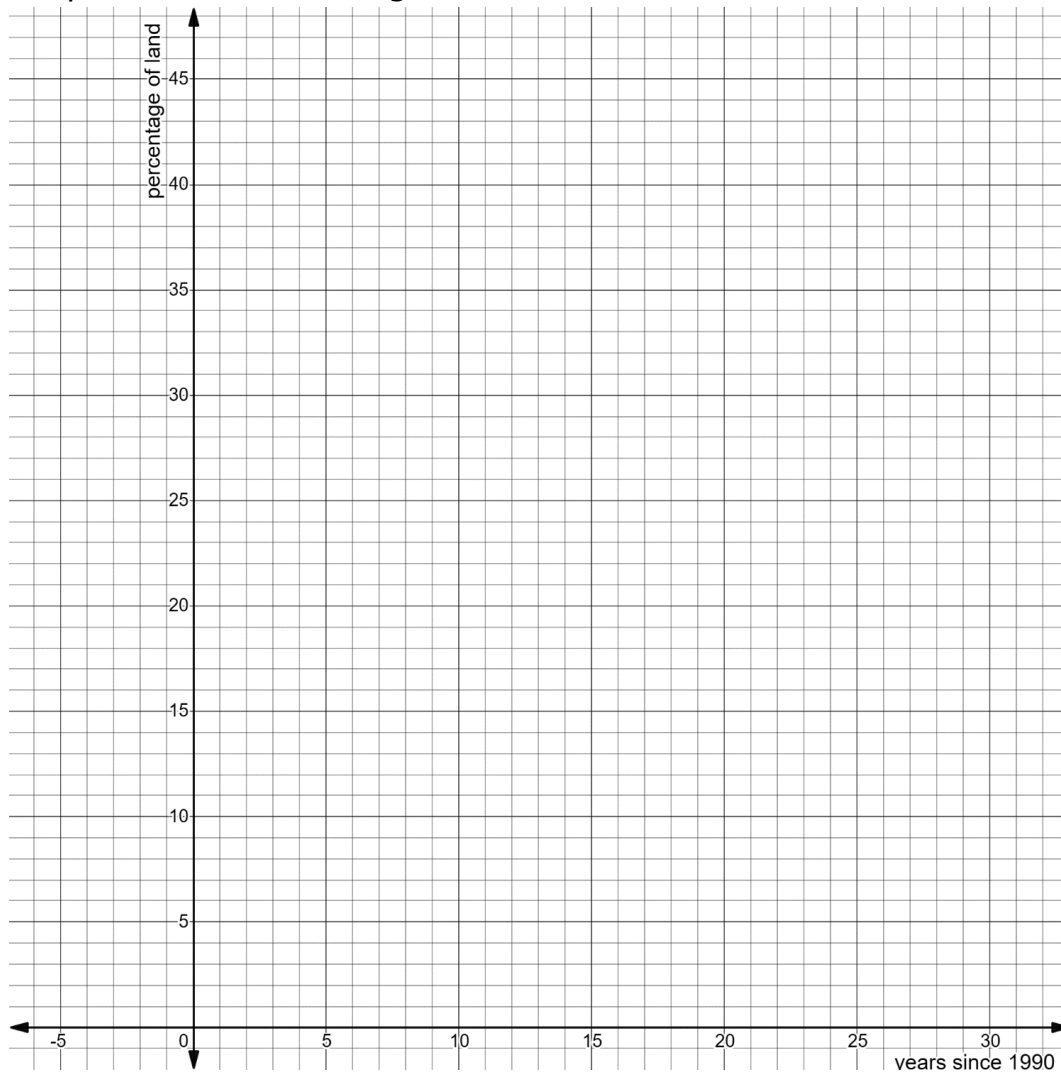
The function $f(x) = -\frac{3}{4}x + 12$ is shown on the coordinate grid.



1. Use the graph to find the solution $f(x) = 18$. Label the point that represents the solution.
2. Solve $-\frac{3}{4}x + 12 = 18$. Show your work.
3. Use the graph to find the solution $f(x) = -6$. Label the point that represents the solution.
4. Solve $-\frac{3}{4}x + 12 = -6$. Show your work.

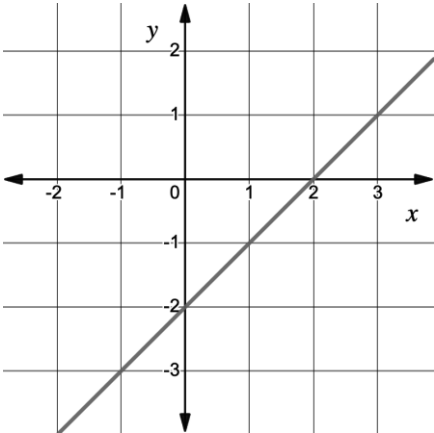
From 1990 to 2020, in Nigeria the percentage of land that are forests has declined. The function $F(t) = -0.2t + 29$ models the percentage of the land, F , that is forested where t is the number of years since 1990.

5. Explain why the values of t should have a constraint.
6. Write the constraint for the values of t using set builder notation.
7. Write the constraint for the values of F using set builder notation.
8. Graph the function using the constraints.



9. Use the graph to find the solution $F(t) = 25$. Label the point that represents the solution.
10. Solve $0.5t + 47 = 55$. Show your work.

For questions 1 through 4, use the table showing Lines B and M .

Line B	Line M
$B(x) = \frac{3}{5}x - 4$	

- Which linear function has the steeper slope?
- Which line has the larger x -intercept?
- Name all the key feature(s) that are the same for both linear functions.
- Create a linear function that has the same y -intercept as Line B .
- Complete the statements below to describe the end behavior of Line B .

a. As $x \rightarrow +\infty$, $y \rightarrow$

- ☐ $-\infty$.
☐ ∞ .

b. As $x \rightarrow -\infty$, $y \rightarrow$

- ☐ $-\infty$.
☐ ∞ .

For questions 6 through 10, use the table showing Lines G and R .

Line G	Line R										
A line passes through the points $(-6, 0)$ and $(4, 15)$.	<table><tr><th>x</th><th>y</th></tr><tr><td>-3</td><td>4</td></tr><tr><td>0</td><td>5</td></tr><tr><td>3</td><td>6</td></tr><tr><td>6</td><td>7</td></tr></table>	x	y	-3	4	0	5	3	6	6	7
x	y										
-3	4										
0	5										
3	6										
6	7										

6. Which linear function has the smaller slope?
7. Which linear function has the larger y -intercept?
8. Name the key feature(s) that are different for the functions.
9. Create a linear function that has the same slope as Line R .
10. Consider the end behavior for the two functions.
 - a. Complete the statement.

Lines G and R have

- ☐ the same end behavior.
 - ☐ different end behaviors.
 - b. Write the end behavior of the functions.

Algebra 1
Unit 4
Lesson 9 Practice

Name _____
Date _____
Period _____

Use the following information to answer questions 1 through 5.

The lines of the equations $y - 7 = 3(x - 2)$ and $-\frac{1}{3}x + y = 1$ are shown on the same coordinate grid.

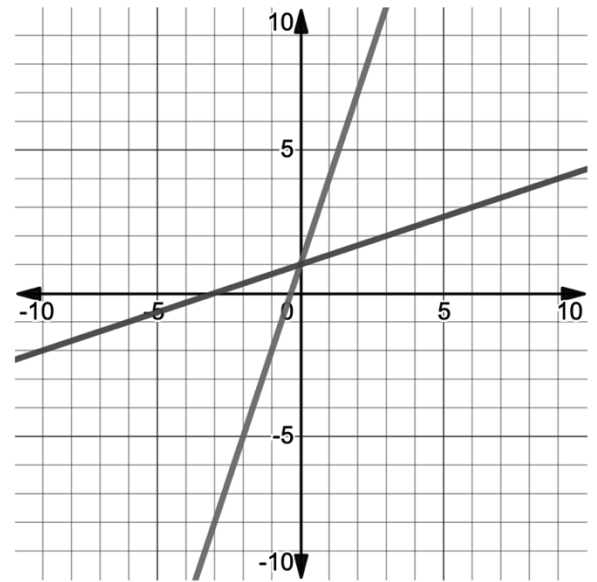
1. What is the slope of the line representing the equation $y - 7 = 3(x - 2)$?

2. What is the slope of the line representing the equation $-\frac{1}{3}x + y = 1$?

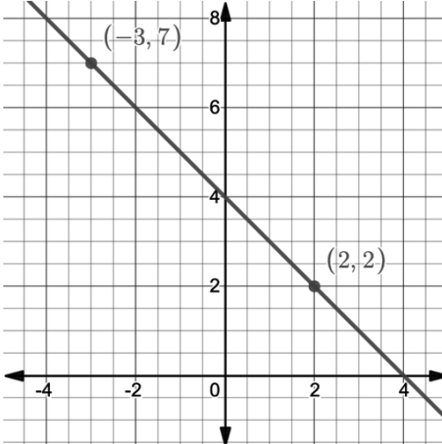
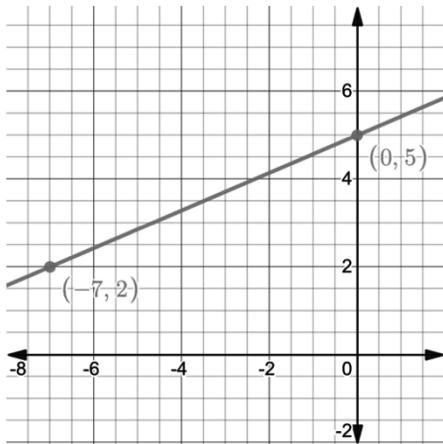
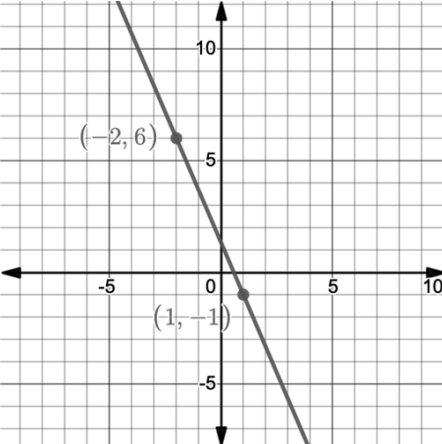
3. What is the slope of a line that is perpendicular to $-\frac{1}{3}x + y = 1$?

4. Explain how to determine if two lines are perpendicular.

5. Determine if the lines are parallel, perpendicular, or neither.



Use the following lines to answer questions 6 through 8.

Line D	Line G	Line K
		
Line M	Line N	Line P
$-x - y = 6$	$y = -x + 8$	$\frac{7}{3}x - y = -1$

6. Which lines are parallel?
7. Which lines are perpendicular?
8. What is the slope of a line that is parallel to Line P?
9. Determine if the lines of the two equations are parallel, perpendicular, or neither.

Equation H	Equation D
$-2x - 4y = -8$	$-2x + 4y = -8$

10. Determine if the lines of the two equations are parallel, perpendicular, or neither.

Equation S	Equation W
$-5x + y = -3$	$y - 8 = 5(x - 3)$

Algebra 1
Unit 4
Lesson 10 Practice

Name _____
Date _____
Period _____

1. Write the equation of the line that passes through $(-2, 1)$ and is parallel to $y = 2x - 3$ in point-slope form.
2. Write the equation of the line that is parallel to $y = 5$ and passes through point $(4, 2)$.
3. Write the equation of the line that contains point $(0, -3)$ and is parallel to $4x + 3y = 6$ in slope-intercept form.
4. Write the equation of the line that is parallel to $x + 6 = 0$ and passes through point $(5, -6)$.
5. Write the equation of the line that passes through point $(3, -4)$ and is parallel to $y + \frac{1}{7} = -\frac{8}{7}(x - 1)$ in standard form.
6. Write the equation of the line, in all three forms, that passes through point $(-4, 11)$ and is parallel to $y + 8 = -\frac{5}{3}(x - 3)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

7. Write the equation of a line that is parallel to $y = 0$ and contains point $(7, 11)$.

8. Line K passes through $(-5, -8)$ and is parallel Line R which has an undefined slope. Write the equation of Line K .

9. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(10, 1)$ and $(6, -1)$ and passes through point $(8, 3)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

10. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(2, 7)$ and $(-1, 17)$ and passes through point $(2, 20)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

Algebra 1
Unit 4
Lesson 11 Practice

Name _____
Date _____
Period _____

1. Write the equation of the line that passes through $(4, 2)$ and is perpendicular to $y = 3x + 1$ in point-slope form.
2. Write the equation in slope-intercept form of the line that is perpendicular to $x = 2$ and passes through point $(2, -1)$.
3. Write the equation of the line that contains point $(-5, 5)$ and is perpendicular to $\frac{5}{9}x - y = 4$ in standard form.
4. Write the equation of the line that is perpendicular to $2x + 5y = 6$ and passes through the origin.
5. Line F passes through $(-12, 0)$ and is perpendicular Line Z which has an undefined slope. What is the slope of line F ?
6. Write the equation of the line that is perpendicular to the line that passes through the points $(7, -3)$ and $(7, 14)$ and passes through point $(4, -\frac{1}{2})$.
7. Write the equation of a line that is perpendicular to $-y = -x$ and contains point $(-1, -1)$.

8. Write the equation of the line, in all three forms, that passes through point $(6, -2)$ and is perpendicular to $y + 3 = -7(x + 1)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

9. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(20, 0)$ and $(0, -10)$ and passes through point $(-10, 10)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

10. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(8, 9)$ and $(-4, -7)$ and passes through point $(0, -5)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

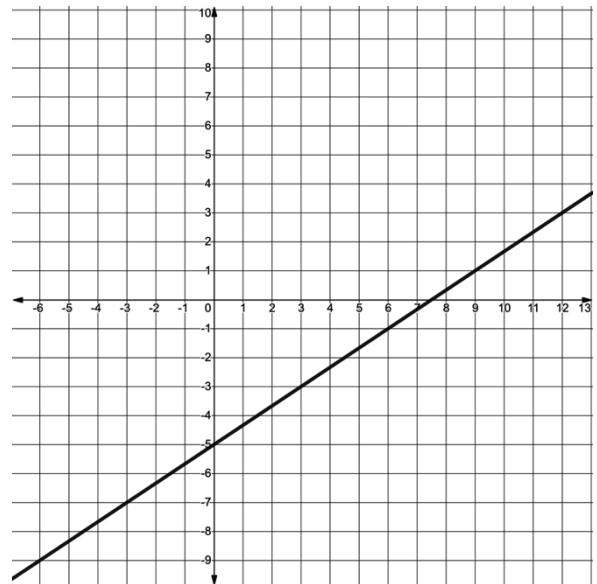
Algebra 1
Unit 4
Lesson 12 Practice

Name _____
Date _____
Period _____

1. Write the equation of the line that passes through $(3, 2)$ and is parallel to $3x - 2y = 1$.

Use the line shown on the coordinate grid to complete questions 2 through 4.

2. What is the slope of the line that is perpendicular to the line shown on the coordinate grid?
3. Write the equation of the line that is perpendicular to the line shown on the coordinate grid and that passes through $(1, 0)$.



4. Write the equation of the line that is parallel to the line shown on the coordinate grid and that passes through $(5, -4)$.
5. Write the equation of the line in standard form that is perpendicular to $y = -\frac{1}{4}x - 13$ and passes through $(1, 7)$.

Use the following information about Line U to complete questions 6 through 8.

Line U passes through $(-2, -2.5)$ and $(1, 0)$.

6. Line Z is parallel to Line U . What is the slope of Line Z ?
7. Line Z passes through $(-6, 9)$. Write the equation of Line Z in slope-intercept form.
8. Line C passes through $(7, 3)$ and is perpendicular to Line U . Write the equation of Line C in point-slope and standard forms.

Point-Slope Form	Standard Form

9. Write the equation of the line that is parallel to $-21x + 18y = 3$ and passes through point $(-6, -3)$ in point-slope form.
10. Write the equation of the line that is perpendicular to $10x = -y$ and passes through point $(4, \frac{2}{5})$.